

Bootstrap Test

Part 1: Multiple Choice Questions (30 Marks)

I. Basics and Setup

Q1. What is the primary purpose of Bootstrap, according to your notes?

- A. To manually design websites using HTML, CSS, and JavaScript.
- B. It is a Component library used to build interactive UI faster.
- C. It is a package manager tool for installing libraries.
- D. To define custom CSS attributes with dot notation.

Q2. In the context of Bootstrap, what does the term "Component Library" refer to?

- A. A library that only contains JavaScript functionality.
- B. A collection of predefined templates (Markup, Styles, and Functionality) that can be implemented and customized.
- C. A licensing agreement that allows free usage.
- D. A tool used exclusively for front-end framework development.

Q3. Which of the following is cited as the latest version of Bootstrap in your notes?

- A. Bootstrap 3
- B. Bootstrap 4
- C. Bootstrap 5
- D. Bootstrap 6

Q4. To use Bootstrap without downloading and installing files via NPM, what is the alternative method mentioned in your notes?

- A. Using a local file path.
- B. Linking the CDN (Content Delivery Network) directly in the `<head>` section of the page.
- C. Installing Visual Studio Code extensions.
- D. Using the `node -v` command.

Q5. A Bootstrap component is said to comprise of which three elements?

- A. Node.js, NPM, and VS Code.
- B. HTML (Markup), CSS (Styles), and JavaScript (Functionality).
- C. CDN, `data-bs-toggle`, and `data-bs-target`.
- D. `.alert`, `.modal`, and `.offcanvas`.

II. Layout and Spacing (Box Model/Containers)

Q6. Which Bootstrap container class is recommended as the "best option for all screens" because it adjusts responsively to any device?

- A. `.container`
- B. `.container-lg`
- C. `.container-fluid`
- D. `.container-xxl`

Q7. What is the correct Bootstrap class utility for applying margin to the *left* and *right* sides of an element?

- A. `.mt-{} , .mb-{}`
- B. `.mx-{}`
- C. `.my-{}`
- D. `.m-{} , .p-{}`

Q8. Which Bootstrap contextual color class corresponds to the color *Red*?

- A. `.primary`
- B. `.success`
- C. `.warning`
- D. `.danger`

Q9. To apply a circular border-radius to an image in Bootstrap, the height and width of the image must be the same. Which class is used for this purpose?

- A. `.rounded-3`
- B. `.rounded-pill`
- C. `.rounded-circle`
- D. `.border-round`

III. Typography and Utilities

Q10. In Bootstrap's font size utility classes (`.fs-{size}`), which size number represents the *highest* font size?

- A. `fs-6`
- B. `fs-3`
- C. `fs-1`
- D. `fs-medium`

Q11. To force a paragraph of text to be displayed in all capital letters, which text transformation utility class should be used?

- A. `.text-capitalize`
- B. `.text-bold`
- C. `.text-uppercase`
- D. `.text-transform-upper`

Q12. The class `.text-decoration-line-through` is used to achieve which CSS property on text?

- A. Bold text
- B. Italic text
- C. Strikethrough text
- D. Underlined text

Q13. To align text to the right side of its container, which class is correct?

- A. `.text-left`
- B. `.text-center`
- C. `.text-end`
- D. `.align-right`

Q14. Which utility class is correct for setting an element's width to 50%?

- A. `.w-50`
- B. `.width-50`
- C. `.w-half`
- D. `.w-75`

Q15. The font weight class `.fw-medium` is used to apply which style?

- A. Extra light font weight.
- B. Bold font weight.
- C. A slightly lighter weight than bold.
- D. The default font weight.

IV. Buttons and Forms

Q16. Which class is the essential base class that must be applied to any element (like `<button>` or `<a>`) to style it as a Bootstrap button?

- A. `.btn-primary`
- B. `.btn-default`
- C. `.btn`
- D. `.button`

Q17. To create a group of buttons stacked vertically, which class should be applied to the wrapper container?

- A. `.btn-group`
- B. `.btn-toolbar`
- C. `.btn-group-vertical`
- D. `.btn-stack`

Q18. What is the primary purpose of using the class `.btn-outline-{contextual}`?

- A. To create a large button.
- B. To create a button with a transparent background and a colored border.
- C. To create a small button.
- D. To style a button as a hyperlink.

Q19. Which form class is specifically designed to style the standard text-based input fields (`<input type="text">`, `<input type="email">`, `<textarea>`)?

- A. `.form-label`
- B. `.form-select`
- C. `.form-control`
- D. `.form-check-input`

Q20. In a professional Bootstrap form structure, the notes recommend keeping every category (like username, password) inside a separate what?

- A. `<p>` tag.
- B. `<span>` tag.
- C. `<div>` tag.
- D. `<table>` tag.

V. Components (Alerts, Modals, Offcanvas)

Q21. The primary difference between a Bootstrap Alert and a Modal is that an Alert is an *embedded message box* that is part of the webpage, while a Modal is a...

- A. Static HTML element.
- B. Hidden element with no dynamic control.
- C. **Pop-up dialog** with content in the background.
- D. Navigation bar component.

Q22. Which HTML attribute is used in Bootstrap components to dynamically close a component (like an Alert or Modal) when clicked?

- A. `data-bs-toggle`
- B. `data-bs-target`
- C. `data-bs-dismiss`
- D. `data-bs-close`

Q23. Which class is used to center a Modal dialog vertically on the screen?

- A. `.modal-dialog`
- B. `.modal-dialog-scrollable`
- C. `.modal-fullscreen`
- D. `.modal-dialog-centered`

Q24. In the context of opening a Modal, the `data-bs-target` attribute must be set to the value of what?

- A. The contextual color class.
- B. The ID of the modal element (`#id`).
- C. The class name of the modal element.
- D. The URL of the modal content.

Q25. Which component, recently introduced in Bootstrap, is similar to a Modal but provides a dialog that can span in any orientation and is often used for side-panel toolbars/nav-bars?

- A. Alert
- B. Offcanvas
- C. Carousel
- D. Dropdown

Q26. If you want to add animation when a Modal opens, which class should be added to the main modal container?

- A. `.animated`
- B. `.fade`
- C. `.slide`
- D. `.dynamic`

Q27. When creating a closable Alert, which class is required alongside `.alert` and `.alert-{contextual}`?

- A. `.alert-close`
- B. `.alert-title`
- C. `.alert-dismissible`
- D. `.alert-subtitle`

Q28. The utility class `d-flex` is primarily used to activate what layout module in a container?

- A. Grid System
- B. Floats
- C. **Flexbox** (Implied by the utility class naming convention, as Flex utilities like `flex-row` and `justify-content` are listed after `.d-flex`).
- D. Position

Q29. What is the role of `data-bs-slide-to` on the Carousel Indicators?

- A. To set the time interval for the slide.
- B. To specify the ID of the Carousel.
- C. To indicate which slide index (0, 1, 2, etc.) should be displayed when the indicator is clicked.
- D. To set the slide's transition animation.

Q30. Which of the following is *not* a valid size value for Bootstrap's Width and Height utility classes (`.w-{size}` or `.h-{size}`)?

- A. 25
 - B. 50
 - C. 100
 - D. 75 (Only 25, 50, 100 are listed as allowed values in your notes).
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Part 2: Real-World Problem Statements (12 Problems)

1. Alert Component

Problem: Show a permanent success message after a user successfully submits a form. The message should contain a "Go to Dashboard" link that is styled as an alert link, and the entire message must be dismissible by a close button.

Required Solution Classes/Attributes:

```
.alert, .alert-success, .alert-dismissible, .alert-link, data-bs-dismiss="alert"
```

2. Modal Component (Login/Registration)

Problem: Create a "Click to Register" button that, when clicked, opens a full-screen registration form pop-up dialog with animation. The form should be scrollable if the content is long.

Required Solution Classes/Attributes (for button and modal):

- **Button:** `data-bs-toggle="modal", data-bs-target="#register-modal-id"`
- **Modal Container:** `.modal, .fade`
- **Modal Dialog:** `.modal-dialog-scrollable, .modal-fullscreen`

3. Offcanvas Component

Problem: Implement a side navigation menu on a mobile site. The menu should appear from the right side of the screen when a button is clicked.

Required Solution Class:

```
.offcanvas-end
```

4. Carousel Component

Problem: Set up a carousel on a homepage. The second slide (`.carousel-item`) should have an automatic transition interval of 3 seconds.

Required Solution Attribute:

```
data-bs-interval="3000" (on the second .carousel-item element)
```